COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-DBS-014: Database Support in Cloud Services

Document ID	TS-DBS-014
Title	Database Support in Cloud Services
Executive Sponsor	Database Services Management, DBS Manager
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT supported agencies; employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
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Purpose

OIT is statutorily mandated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy the stakeholder needs surrounding OIT supported databases and their versions, the following is submitted as the standard for specifications, methodologies and practices to provide the service.

The purpose of this document is to communicate the standards that the Governor's Office of Information Technology (OIT) has developed with the intent of providing enterprise standard configurations for Database Support in Cloud Services. Standards are based on a best-of-value decision assessment involving multifarious factors, such as stakeholder needs, hardware configurations that are serviced and supported by OIT, as well as operational and security best practices guidelines.

The ultimate purpose of the Database Support in Cloud Services is to describe the minimum requirements for designing, installing, securing, monitoring, maintaining, protecting, and decommissioning any OIT supported Database environment.

Scope

These standards apply to OIT and OIT supported agencies. This Standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Additionally, the standards assigned and prescribed by this document, instituted for use by the approval of the OIT Architecture Review Board (ARB), should be used to govern purchase and deployment decisions deemed as “go-forward” or “net-new” strategies or implementations, and are

not intended for use as retroactively enforceable mandate or policy for immediate conformance (especially in cases where cost implications may be prohibitive).

Any/all standards are predicated with the assumption that there can be no point of enforcement unless and until there is an effort to re-procure, renew (contracts), reconfigure (including upgrades), or redevelop the system. At any time that a decision is made to make an investment of resources in sustaining or improving the system, conformance to existing standards should be part of the assessed level of effort for that investment activity. For guidance (or enforcement) in conformance to standards during any of these types of activities, an email request may be sent to OIT_OEA@state.co.us with the subject line "Request for Architecture Review: <projectName/systemName>"

For the purpose of enforcement, this standard is rated CAT 1 according to the Enforcement Category Chart.

Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Update effective date and/or last review date.
- Publish the ARB approved Standard to the OIT public facing website.

Architecture Review Board (ARB)

- Engage stakeholders and subject matter experts to determine standard.
- Review/Approve Standard (this document).

Database Services Management - DBS Manager

- Owner of Standard.
- Engage stakeholders and subject matter experts to determine standard.
- Update document standards and version number.
- Present to the Architecture Review Board for approval and adoption.

Additional Assignments

Database Services Management

- Responsible for enforcement and approval of changes of policies, standards, guidelines, procedures.

Database Services Database Administrator (DBA)

- Responsible for knowing and adhering to policies, standards, guidelines, procedures as part of job description.

All OIT staff are responsible to be aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

Standards

Goals

- Ensure Database Services is able to provide support, patching, upgrades and installations specific to database platforms, versions, and environments in the Cloud environment.

Process

- Cloud database operations, support and maintenance including any database categorized as Database as a Service (DBaaS) must meet the following criteria:
    - Any database request for installation, upgrade or maintenance must be initiated by Database Services.
    - The Database Services team will manage, support, secure, and optimize database operations in the cloud environment.
    - Any vendor supported cloud database must comply with the OIT Database Services standards.
- Installing and upgrading database platforms and tools includes but is not limited to infrastructure as code to build any database instance using terraform, ansible or any other automation tool, subject to the following rules:
    - If a DevOps pipeline is in use to build out the database infrastructure, at least two members from the database services team must approve the change before any deployments can proceed.
    - If the installation is done through the console or CLI via infrastructure as code i.e. Terraform, the database services team must approve Azure DevOps Pull Requests or console work before being implemented.
- Initial system storage and capacity planning requirements must follow this rule:
    - Allocating initial system storage, capacity planning and future storage requirements for the database system as per the sizing requirements of the agency must be approved by Database Services.
- Database Patching guidelines are defined below:
    - The scheduling, testing and applying of database patches will be performed by database services.
- Database users and security maintenance regulations are outlined below:
    - Database users and security maintenance will be done by the database services team including the creation and maintenance of any RDS cloud administrator accounts. The master account and password can not be given to any individual outside of the Database Services Team.
    - Controlling, monitoring, auditing user access, and enforcing password requirements to the database will be done by using least privilege as per the requirements listed in TS-DBS-006: Database Security. Password rotation should be used when possible as defined in the CISP.
- Supported Database Versions must conform to enterprise standards:
    - Any cloud database supported by Database Services must be in the approved supported database versions listed in TS-DBS-001 Supported Databases and Versions.
- Database Related Services are managed as follows:
    - Any database related services in the Cloud such as DMS etc. will be initiated and managed by Database Services and initial provisioning will be done through Terraform.
    - The creation and support of database maintenance jobs will be managed by Database Services.
- Database Backup and Recovery operational standards include:
    - Backing up, restoring and maintaining database backups is the responsibility of the Database Services team.

    - If using an RDS or a managed instance, all databases in the instances will be restored regardless of the number of databases.
- Cloud Database Inventory tracking operations require:
    - Database services will update and maintain inventories for all OIT managed databases in the cloud infrastructure.
- Database User Administration guidelines are governed by separate authority:
    - Database user creation and maintenance is defined in the database standard TS-DBS-008: Database User Administration.
- Additional Support and Services provisions consist of:
    - Monitoring and optimizing database performance by using and maintaining any cloud provided tools such as performance insight and cloud watch.
    - Maintaining archived data.
    - Ensuring compliance with database vendor license agreements.
    - Contacting database vendors for technical support (if required). This would include support accounts with any Cloud provider.
    - Incident triage.

Requirements

- Cloud technology infrastructure choices must align with organizational requirements:
    - Database Services will be involved in Cloud selection and database platform preferences.

Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of and approved by the "Owner of Standard" listed in Section 3 (Roles and Responsibilities) of this document.

Requests may be acquired by completing the "EXCEPTION REQUEST to Technical Standard" form and including the completed form (including all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

References

OIT documents and policies

- State of Colorado Style Library
- TS-OEA-001: Technology Accessibility for Persons with Disabilities
- Enforcement Category Chart
- Colorado Information Security Policies (CISPs)

State documents and policies

- Colorado Brand Guidelines
- C.R.S. 24-37.5-101, et seq.
- OIT DBS Database Software Management Policy

Third party resources

- My Oracle Support

- Microsoft Support
- AWS Support (access through the AWS account)

Definitions

Standard: A rule that specifies a particular course of action or response to a given situation. Standards are mandatory directives to carry out management's policies and are used to measure compliance with policies. Standards serve as specifications for the implementation of policies and are designed to promote implementation of high-level organization policy rather than to create new policy in themselves.

Production Database: State Agency database used for production purpose to conduct State business.

Non-Production Database: State Agency database used for non-production purpose such as development, testing, staging, demonstration or training.

Business: Employees of supported State agencies, and their respective stakeholders that use DBS supported databases.

EA: Enterprise Application development group responsible for application development.

CLI: Command line interface

Database Instance: Any isolated database provided in cloud services that are available by AWS, Microsoft Azure, and the Google platform.

OEA (Office of Enterprise Architecture): The Office of Enterprise Architecture was formed as an office under OIT by Governor's Rule 8 CCR 1501-7.

OIT (Office of Information Technology): The Governor's Office of Information Technology was created and governed by C.R.S. 24-37.5, et seq.

Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification for granting the exception. Typically an exception request is subject to review and approval by an authority in the area.